

# Candlelight - User Manual

## 1. Introduction

Candlelight is a simple roguelike dungeon-explorer game, where you play as a candle, using your flames to defeat enemies and rescue dowsed friends! The game's core mechanic is Wax, which functions not only as your health, but as your "mana" bar for casting flames, much like how the brighter a candle burns, the shorter it gets. The player must juggle their health and mana - is it worth running in to clear the next wave, or should they retreat and recollect?

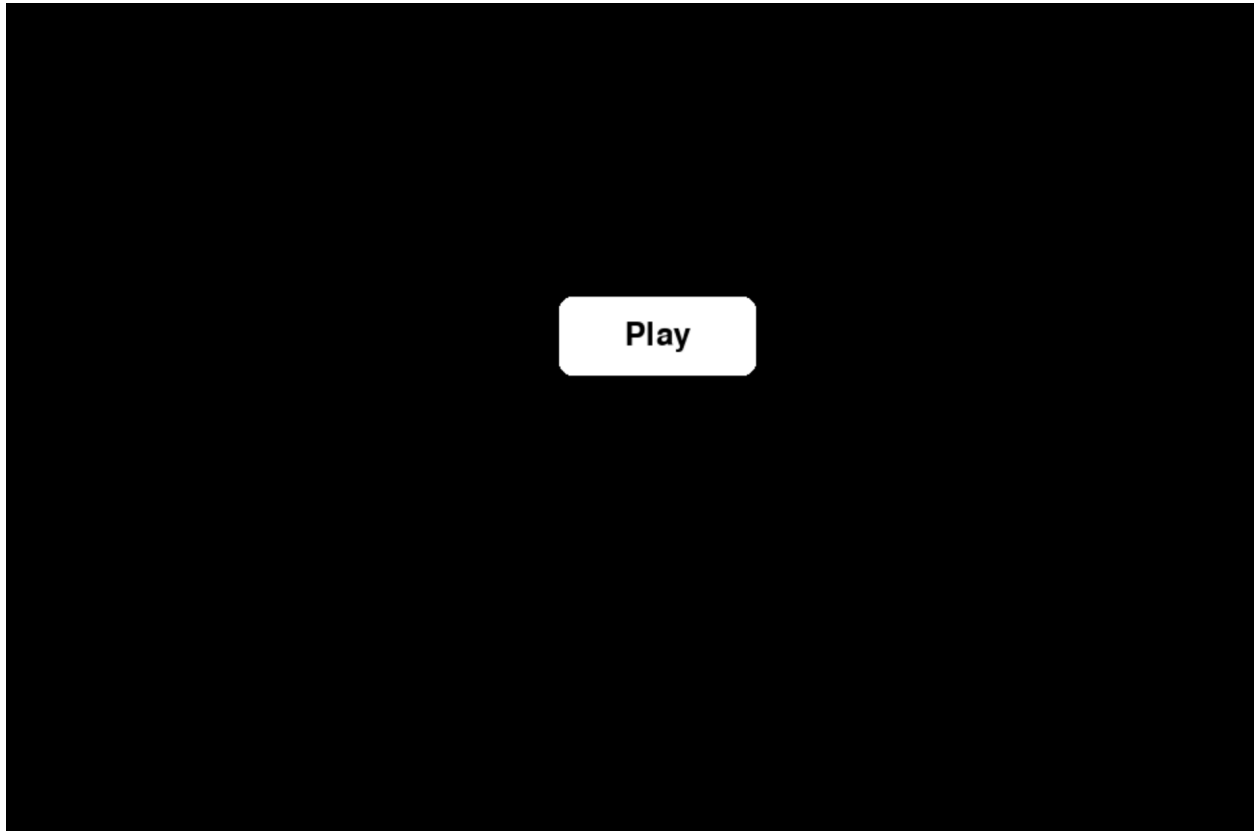


fig. 1 - Start menu

## 2. System Requirements

**Operating system:** Windows 10+, macOS 11+, or Linux

**Python:** 3.10 or later

**Libraries:** pygame 2.6.0

**Display:** minimum 960×640

**Storage:** under 10 MB

**Input:** keyboard required

### 3. Installation

1. Install Python 3.10+ from [python.org](https://python.org)
2. Open a terminal in the project folder
3. Install dependencies: `pip install -r requirements.txt`
4. Run the game from `main.py`
5. Note that `candlelight.db` will be created automatically on first run to store your progress
6. Also note that `dungeon.py` is legacy and not required to run Candlelight

### 4. Controls

| Key           | Action                                            |
|---------------|---------------------------------------------------|
| W / A / S / D | Move character up, left, down, right respectively |
| Arrow keys    | Cast flame in chosen direction                    |
| ESC           | Pause / Resume & view controls                    |
| SPACE         | Return to menu from game over screen              |

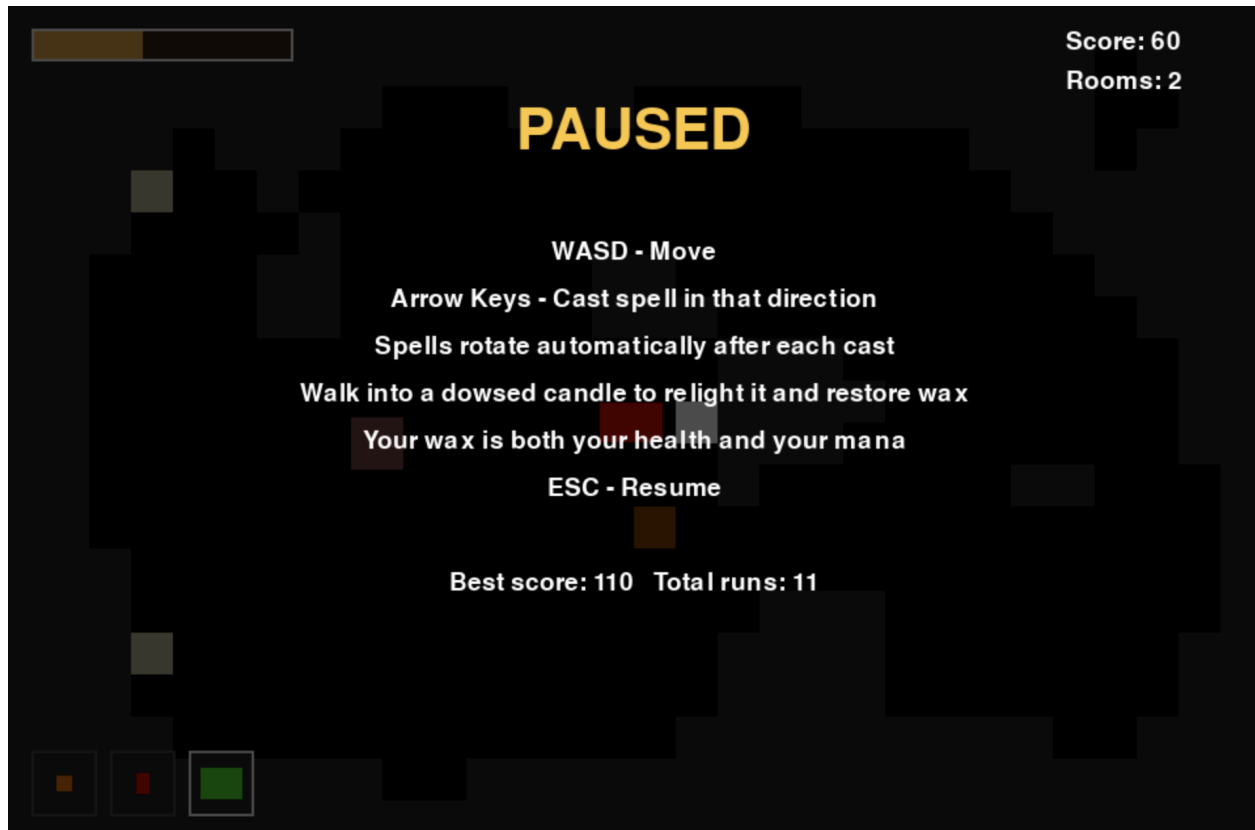


fig. 2 - Pause menu

## 5. How to Play

### 5.1 - Starting a Run

Hit "Play" on the start menu (see fig. 1)

### 5.2 - Moving and Exploring

Use WASD to control your character. Move through the randomly generated caves, and use the brown doors to advance to the next stage

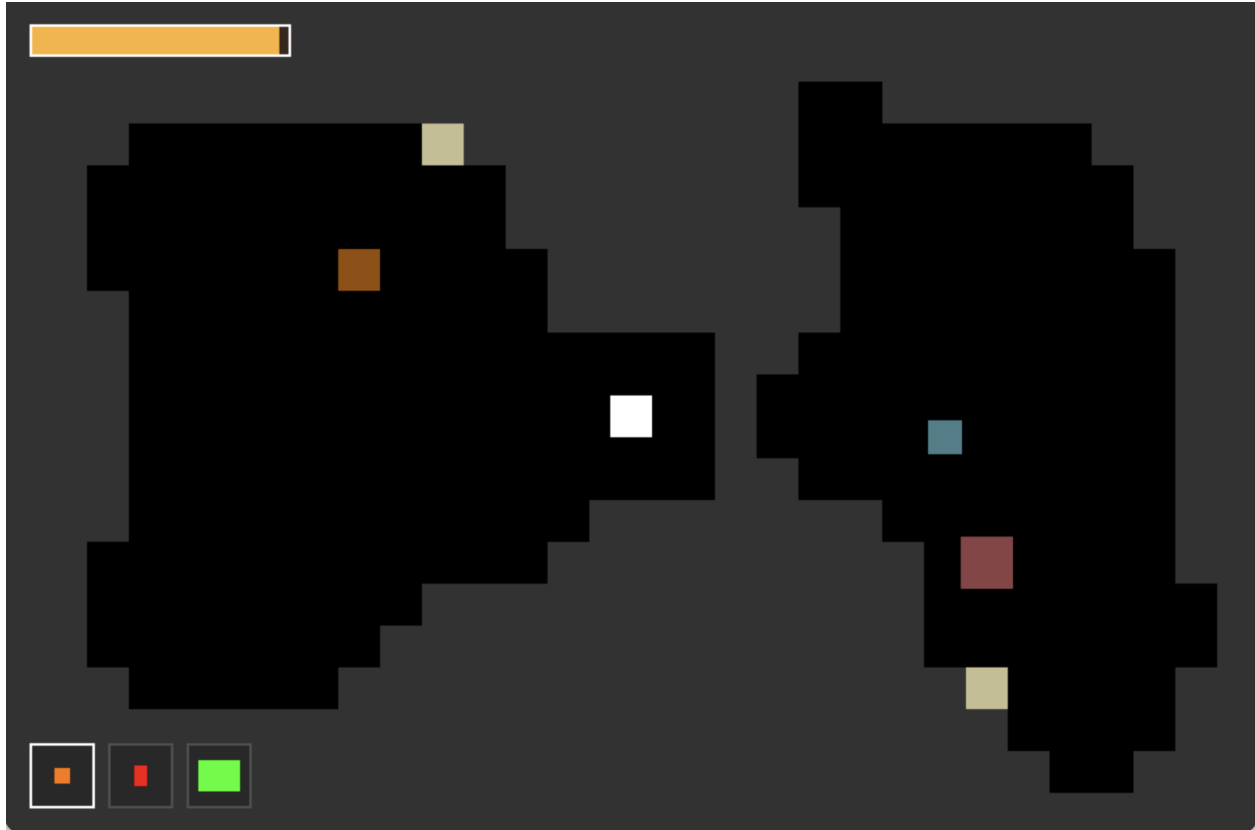


fig. 3 - Player (white) exploring cave

### 5.3 - Wax

The player's Wax functions as both a health and mana mechanic. The "wax-bar" is visible in the top left of the gameplay window (see fig. 3). It will drain slowly over time, and will also be drained when the player casts spells, or is hit by an enemy spell. Once the player's wax hits zero, the run will be ended.

### 5.4 - Casting Spells

The player's arsenal consists of three spells, rotating through a queue. The next spell is displayed in the bottom left corner, with an icon displaying all three selected spells. After each cast, the spell queue will cycle automatically, from left to right. Each spell has different attributes - they are visually differentiated by size and colour, and each cost different amounts of wax to cast.

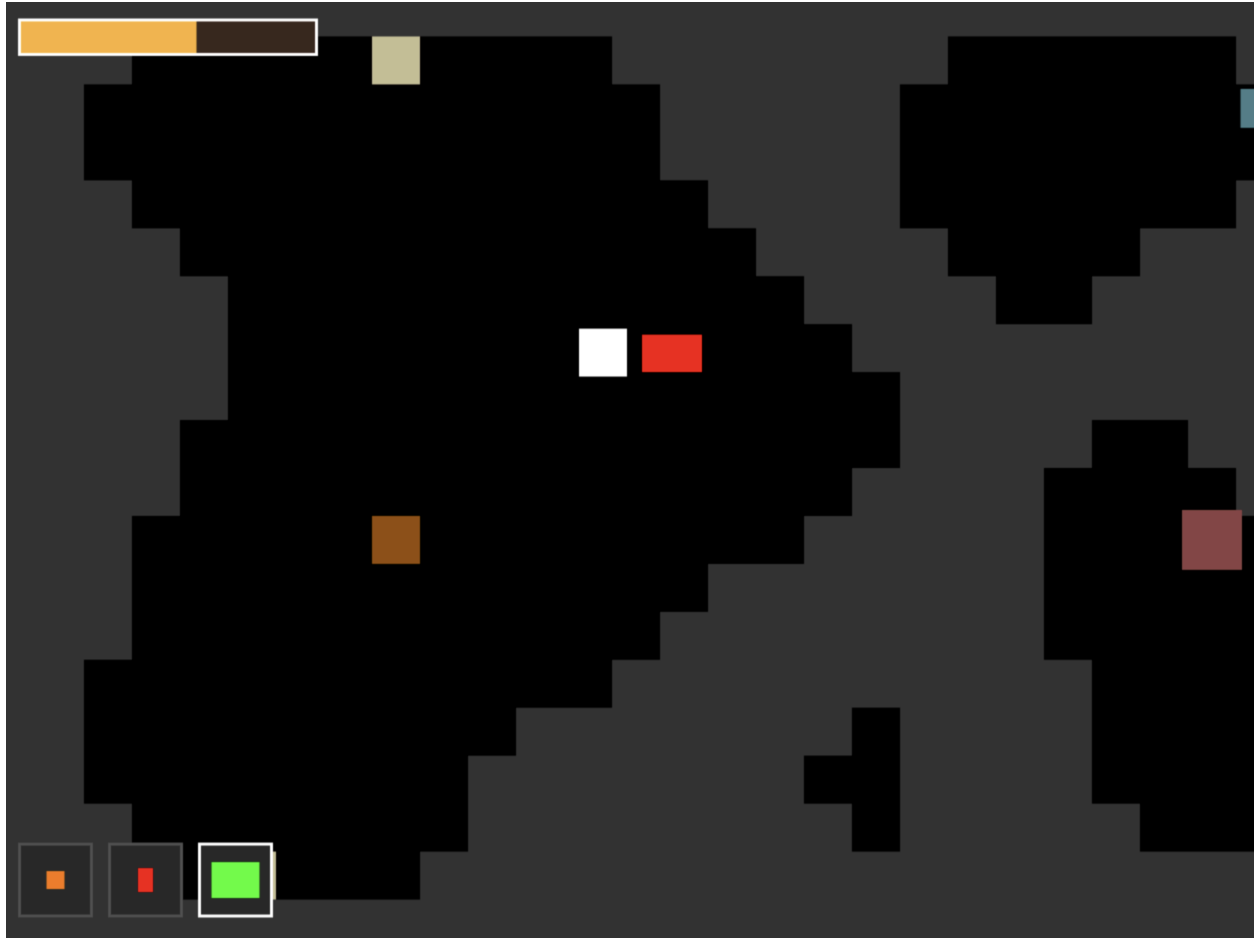


fig. 4 - Player casting spell #2

### 5.5 - Relighting Candles

The player may find dowsed candles scattered across the dungeons. By walking into the candles, the player will relight them and they will change colours (see contrast between fig. 3 and fig. 5). After lighting the candle, the player will receive a bonus amount of wax to replenish their wax bar.

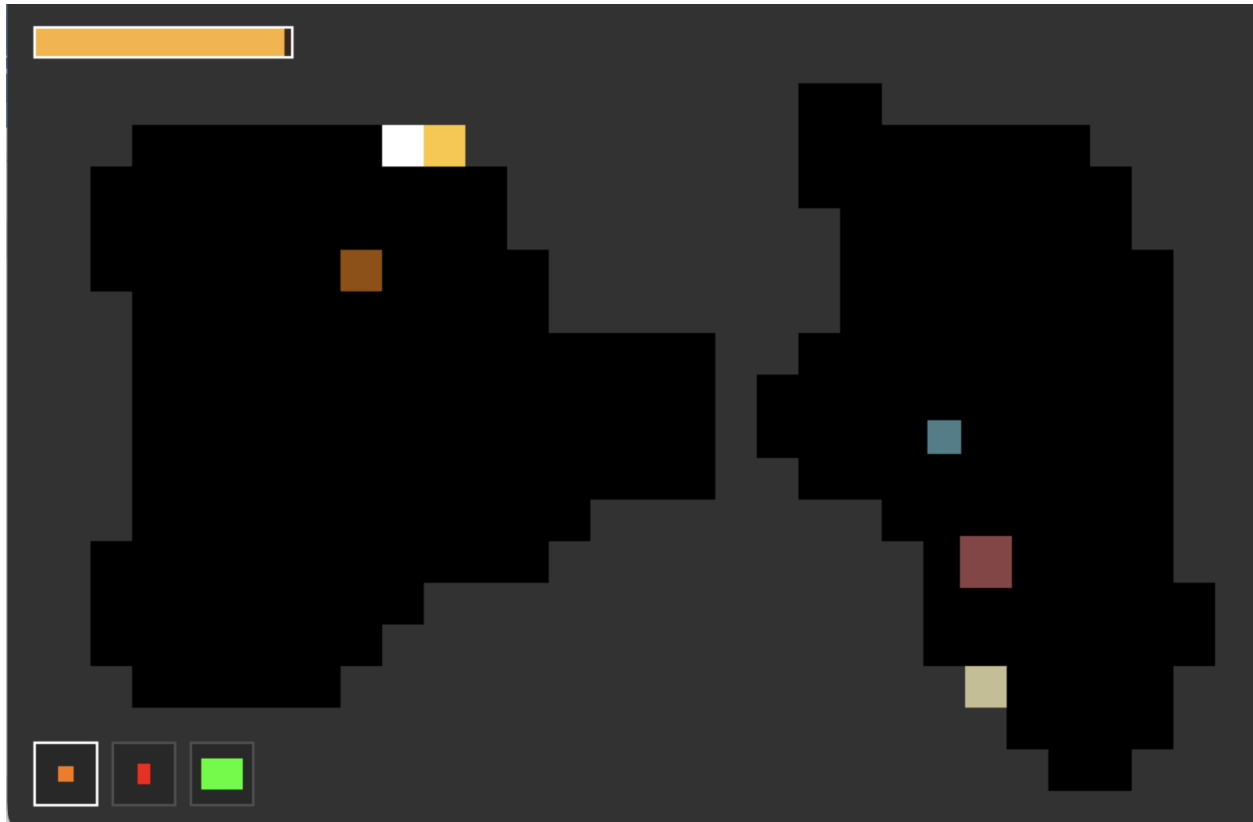


fig. 5 - Player has lit a dowsed candle and received bonus wax

## 5.6 - Enemies

The player will encounter three different types of enemies in the dungeons. These are the wisp, shade and brute. These enemies differ in speed, damage, size and colour, being slow - fast and low - high damaging respectively. The enemies will hunt the player through the cave, and attack at short range. Given the differences in size, the brute will be the least manoeuvrable, and the player may escape its high damage by darting through small cave openings or around tight corners that the brute can't navigate through. By contrast, the player will have a hard time escaping the wisp, and they will likely be better off simply picking them off. Additionally, as the player progresses through the dungeon, larger and larger waves of enemies will spawn - +1 per two levels to be precise, capping at a maximum of 8. See how long you can last against the horde!

fig. 6 - a Brute enemy is attacking the player (see red attack rectangle)



### 5.7 - Clearing Rooms

Each dungeon room contains a number of enemies. The exit door remains locked - shown in dark brown - until every enemy in the room has been defeated. Once the room is clear, the door lightens to indicate it can be entered, and walking into it advances you to the next room.

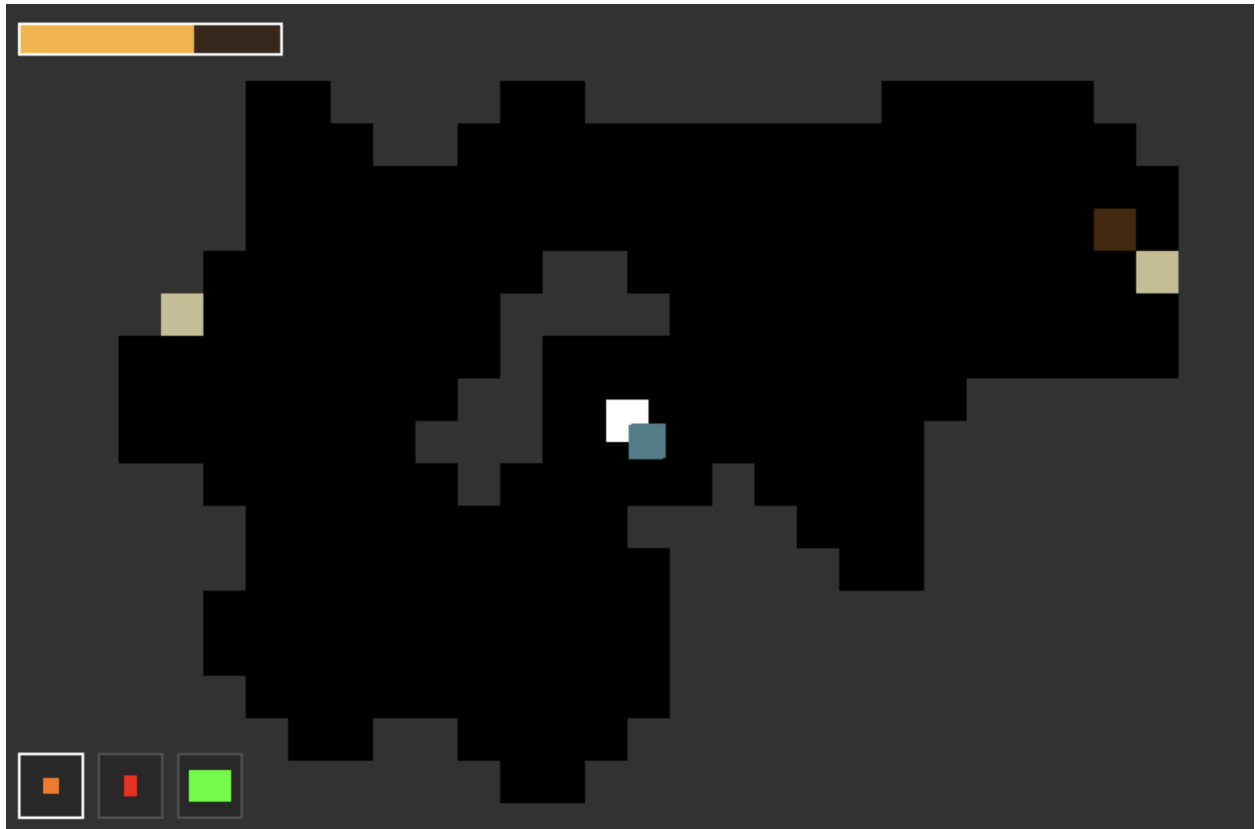


fig. 7 - door (top right) is a deeper brown than usual

### 5.8 - Ending a run

On depletion of the player's wax, the run will end. On player death, they will be redirected to an EXTINGUISHED page, displaying the player's final score, along with amount of rooms cleared and a prompt to return to the start menu. On death, some player stats will persist - for example high scores will be stored along with the player's total number of runs across sessions (see fig. 2). On new game, a new dungeon will be generated and the player will start from zero.



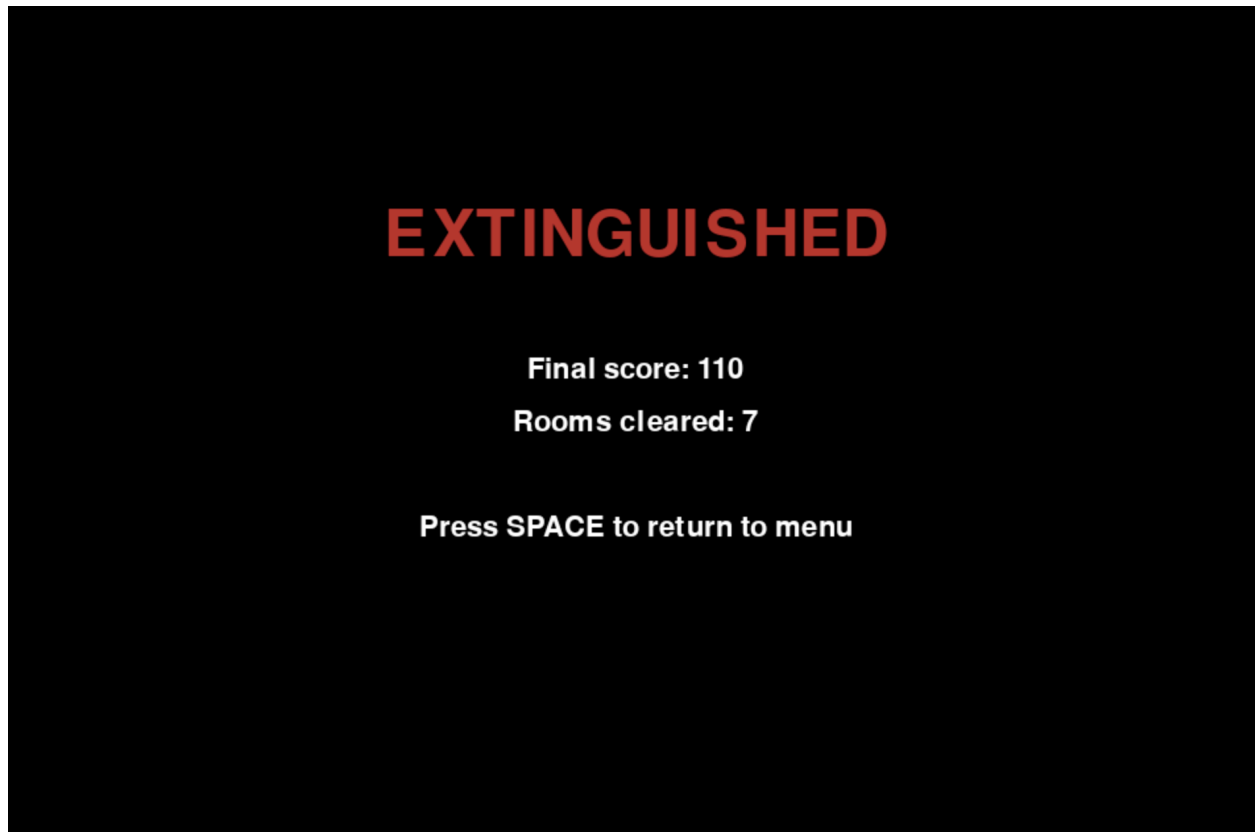


fig. 8 - the game over, or “Extinguished” screen

## 6. Troubleshooting

| Problem                                              | Fix                                                                                                                                                                                                                                                                                                                                                                                                                        |
|------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>ModuleNotFoundError: No module named 'pygame'</i> | Run <i>pip install pygame</i> . If you have multiple versions of python installed, try <i>pip3 install pygame</i> .                                                                                                                                                                                                                                                                                                        |
| Game window doesn't open                             | Confirm you're running python 3.10+ with <i>python -- version</i> .                                                                                                                                                                                                                                                                                                                                                        |
| Game runs slowly                                     | Close other applications. The game is fixed at 60 FPS, and is CPU bound during dungeon generation. Additionally, each enemy recalculates its path using A* twice per second. Performance was tested with up to 1000 enemies; a small first-load delay may be incurred around 50-100 enemies, but the game only begins to slow down towards 1000 concurrent entities. Lower counts may reduce frame rate on older hardware. |

|                               |                                                                                                                                       |
|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| Progress isn't saving         | Ensure the game folder is writable - <i>candlelight.db</i> must exist within the same directory as <i>main.py</i> .                   |
| Deleted <i>candlelight.db</i> | This resets all saved statistics - the game will recreate the file on next launch.                                                    |
| Enemy appears stuck on a wall | Larger enemies cannot fit through single-tile corridors or tighter corners. This is intentional - use narrow passages to escape them. |